2019/20 F78AA Exam Solutions

1. (a) $(1+i)(1-d_{12})^{12} = 1$

$$i = \frac{1}{(1-0.005)^{12}} - 1 = 0.061996$$

(b) $(1+i_{12})^{12}(1-d_6)^6 = 1$

$$d_6 = 1 - \frac{1}{(1+0.005)^6} = 0.029482$$

2. $i = \sqrt{1.1025} - 1 = 0.05$ per half-year

$$\text{PV} = 500a_{\overline{8}|}^{0.05}(1.05)^{-24} + 500a_{\overline{12}|}^{0.04}(1.05)^{-32}$$

PV = 1002.0174 + 984.8047 = 1986.82

3. $$\text{FV} = 100(1.04)^{\frac{7}{12}}(1.025)^4(1.01)^{12} - 85(1.04)^{\frac{11}{3}}(1.025)^{12} - 25(1.01)^3$$

FV = 127.25887 – 104.85672 – 25.75753

= –3.35538 = –3.36

4. $$65.85 = (1-0.25)4a_{\overline{15}|}^{i} + \left[98 - (98-65.85)0.4\right]v^{15}$$

$$65.85 = 3a_{\overline{15}|}^{i} + 85.14v^{15}$$

Let, $f(i) = 3a_{\overline{15}|}^{i} + 85.14v^{15} - 65.85$

Initial Guess: $3\left(\frac{1-(1-15i)}{i}\right) + 85.14(1-15i) - 65.85 = 0 \quad \Rightarrow \quad i \approx 0.05$

$f(0.05) = 6.242770 \quad > 0$

$f(0.06) = -1.187306 \quad < 0$

Linear Interpolation,

$$i^* = 0.05 + 0.01 \times \frac{6.242770}{6.242770 + 1.187306} = 0.0584$$

Annual rate = $1.0584^2 - 1 = 0.1202$

5. (a) Working in millions,
Let, MWRR = i

$$100(1+i)^2+1(1+i)^{\frac{23}{12}}-2.5(1+i)-1(1+i)^{\frac{7}{12}}+5(1+i)^{\frac{1}{12}}=112$$

Initial Guess:

$$100(1+2i)+\left(1+\frac{23}{12}i\right)-2.5(1+i)-\left(1+\frac{7}{12}i\right)+5\left(1+\frac{1}{12}i\right)=112$$

$$i\approx 0.048$$

$$\text{Let, } f(i)=100(1+i)^2+1(1+i)^{\frac{23}{12}}-2.5(1+i)-1(1+i)^{\frac{7}{12}}+5(1+i)^{\frac{1}{12}}-112$$

$$f(0.050)=0.714527 \quad >0$$

$$f(0.045)=-0.329608 \quad <0$$

Linear Interpolation,

$$i^*=0.045+0.005\times\frac{0.329608}{0.329608+0.714527}=0.04658$$

(b) Let, TWRR = j

$$(1+j)^2=\frac{101}{100}\times\frac{106}{102}\times\frac{104}{103.5}\times\frac{105}{103}\times\frac{112}{110}$$

$$(1+j)^2=1.094706$$

$$j=0.04628$$

6. (a) Working in '000.
Let, i = 0.005 per month
To find the DPP, we need the smallest k such that

$$25s_{\overline{k}|}^{0.005}\geq 1000\left(1.005^k\right)$$

$$25\left(\frac{1.005^k-1}{0.005}\right)\geq 1000\left(1.005^k\right)$$

$$1.005^k\geq 1.25$$

$$k\geq\frac{\ln 1.25}{\ln 1.005}=44.74$$

DPP = 45 months.

(b) Profit at the end of the 45th month

$$=25s_{\overline{45}|}^{0.005}-1000\left(1.005^{45}\right)=6.483283$$

Profit at the end of the 60th month

$$=6.483283\times 1.004^{15}+25s_{\overline{15}|}^{0.004}+7000-5000\times 1.004^{60}$$

$$=6.88336+385.68420+7000-6353.20359$$

$$=1039.36397$$

Profit = £1, 039, 363.97

7. (a)

$$50000 = Xv^{\frac{1}{12}} + Xv^{\frac{2}{12}} + \ldots + Xv^{\frac{12}{12}}$$
$$+2Xv^{1\frac{1}{12}} + 2Xv^{1\frac{2}{12}} + \ldots + 2Xv^{1\frac{12}{12}}$$
$$\vdots$$
$$+10Xv^{9\frac{1}{12}} + 10Xv^{9\frac{2}{12}} + \ldots + 10Xv^{9\frac{12}{12}}$$
$$50000 = Xv^{\frac{1}{12}}\left[1 + 2v + 3v^2 + \ldots + 10v^9\right]$$
$$+Xv^{\frac{2}{12}}\left[1 + 2v + 3v^2 + \ldots + 10v^9\right]$$
$$\vdots$$
$$+Xv^{\frac{12}{12}}\left[1 + 2v + 3v^2 + \ldots + 10v^9\right]$$
$$50000 = X\left[v^{\frac{1}{12}} + v^{\frac{2}{12}} + \cdots + v^{\frac{12}{12}}\right]\left[1 + 2v + 3v^2 + \ldots + 10v^9\right]$$
$$50000 = X\left[a_{\overline{12}|}^{\,j}\right]\left[\left(I\ddot{a}\right)_{\overline{10}|}^{\,i}\right]$$

where $i = 0.09$ AND $j = 1.09^{\frac{1}{12}} - 1 = 0.007207$

$m = 12$ AND $n = 10$

(b)

$$a_{\overline{12}|}^{\,j} = 11.456261$$
$$\left(I\ddot{a}\right)_{\overline{10}|}^{\,i} = 33.561570$$
$$X \square 130$$

8. (a) Working in ‘000.

Let, X = First annual repayment

$$500 = X\left[v + 1.05v^2 + 1.05^2v^3 + \cdots + 1.05^{29}v^{30}\right]$$

$$500 \times 1.05 = X\left[(1.05v) + (1.05v)^2 + (1.05v)^3 + \cdots + (1.05v)^{30}\right]$$

$$525 = Xa_{\overline{30}|}^{j}$$

where $1 + j = \dfrac{1.1}{1.05} \quad \Rightarrow \quad j = 0.047619$

$$X = 33.23073$$

First annual repayment is £33,230.73

(b) Let, L_n = the capital outstanding immediately after the nth repayment

$$L_{15} = X\left[1.05^{15}v + 1.05^{16}v^2 + \cdots + 1.05^{29}v^{15}\right]$$

$$L_{15} = X\left(1.05^{14}\right)\left[(1.05v) + (1.05v)^2 + \cdots + (1.05v)^{15}\right]$$

$$L_{15} = X\left(1.05^{14}\right)a_{\overline{15}|}^{j}$$

$$L_{15} = 694050.04$$

Working in ‘000.

Let, Y = Sixteenth annual repayment

$$L_{15} = Ya_{\overline{15}|}^{0.12} + 2500\left(Ia\right)_{\overline{14}|}^{0.12}$$

- $a_{\overline{15}|}^{0.12} = \dfrac{1 - 1.12^{-15}}{0.12} = 6.810864$
- $\left(Ia\right)_{\overline{14}|}^{0.12} = \dfrac{1.12 \times \dfrac{1 - 1.12^{-14}}{0.12} - 14 \times 1.12^{-14}}{0.12} = 37.990592$

$$Y = \frac{694050.04 - 2500\left(37.990592\right) \times 1.12^{-1}}{6.810864}$$

$$Y = 89452.62$$